Observable Identity and Exact Simulation

What Information Preserves, and What It Does Not Determine

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Abstract

A state can be identified by its responses to all admissible future experiments. This yields a canonical behavioral quotient, but does not identify an underlying hidden realization. We give elementary proofs of this distinction, show that exact simulation preserves event probabilities, and separate preservation of information from dynamical invariance and indefinite continuation. A Gaussian fixed point of the Fourier transform illustrates representation invariance without introducing a singularity. These are standard mathematical constructions, presented as a self-contained synthesis rather than a claim of new results.

1 Identity relative to admissible experiments

Let $X \neq \varnothing$ be a set of states, $A \neq \varnothing$ a set of actions, $F_a : X \to X$ a total transition for each $a \in A$, and $O : X \to Y$ an observation map. For a finite word $w = a_1 \cdots a_n$, put $F_w = F_{a_n} \circ \cdots \circ F_{a_1}$; the empty word acts as the identity. Define the complete response profile

$$R(x) = \big(O(F_w x)\big)_{w \in A^*}, \qquad x \sim x' \iff R(x) = R(x').$$

The questions in this definition are precisely the experiments allowed by A, not every conceivable physical measurement.

Theorem 1 (Observable identity)**.** *The quotient $M = X/\!\sim$ is in bijection with* $\operatorname{im} R$*. Its transitions and observation are well defined by*

$$\overline{F}_a([x]) = [F_a x], \qquad \overline{O}([x]) = O(x).$$

Consequently M reproduces every finite observation transcript under every deterministic policy based on previous observations and actions.

Proof. The assignment $[x] \mapsto R(x)$ is well defined and bijective onto its image by the definition of $\sim$. If $x \sim x'$, applying any action a followed by any word w gives equal observations, hence $F_a x \sim F_a x'$. The empty word ensures $O(x) = O(x')$. These facts define the quotient maps. Under an adaptive policy, equal histories select the same next action; induction gives equal subsequent observations. □

Thus each quotient state has one distinct response profile. This does not say that R is injective on X. It is injective precisely when the experiments separate every pair of distinct underlying states.

Proposition 2 (Hidden extensions)**.** *Let $E \neq \varnothing$ and choose arbitrary maps $G_a : X \times E \to E$. The extended system*

$$\widetilde{F}_a(x, e) = (F_a x, G_a(x, e)), \qquad \widetilde{O}(x, e) = O(x)$$

has, from (x, e), exactly the same transcripts as the original system from x, under all the above policies.

Proof. Projection onto X commutes with every transition and preserves observations. Induction proves the assertion, including adaptive actions. □

The observed profiles therefore do not determine the hidden set E or its dynamics. Observable equivalence is not a proof that two realizations are literally the same object.

2 Exact simulation preserves the specified law

Let $(\Omega, \mathcal{F}, P)$ be a prescribed trajectory probability space, $(\mathcal{U}, \mathcal{G}, \mu)$ a seed space, and $D : \mathcal{U} \to \Omega$ a measurable deterministic decoder. Exactness means $D_{\#}\mu = P$.

Theorem 3 (Event preservation)**.** *For every measurable event $B \subseteq \Omega$,*

$$\boxed{\mu(D^{-1}(B)) = P(B).}$$

In particular, a positive-probability event cannot be removed while preserving the same exact trajectory law.

Proof. This is the defining identity of the pushforward measure $D_{\#}\mu$. □

For controlled systems, this identity is imposed for each policy using the same seed law and a decoder consistent at shared histories. It remains an identity for each fixed policy; it does not allow the simulator to change that policy's prescribed probabilities.

For a discrete seed U, Shannon entropy is measured in bits:

$$H(U) = -\sum_{u} \mu(u) \log_2 \mu(u).$$

Minimizing this quantity over exact decoders changes the representation, not the specified law. Two models with identical admissible policy transcript laws and the same output interface impose identical constraints on causal decoders; their feasible seed laws and corresponding entropy optima coincide.

Remark 4 (One source is not a finite information budget)**.** *A uniform seed on $[0, 1)$ can encode infinitely many independent fair bits in its binary expansion. It is one random variable, but it is not a finite-entropy discrete seed. Differential entropy is not a replacement for the discrete seed cost above.*

3 Dynamical invariance is an additional condition

Let $B \subseteq X$ be a designated forbidden set.

Theorem 5 (Invariant-region criterion)**.** *Suppose $x_0 \in S \subseteq X$, $S \cap B = \varnothing$, and*

$$F_a(S) \subseteq S \qquad \text{for every } a \in A.$$

Then every finite action sequence, including every sequence selected by an adaptive policy, stays outside B.

Proof. The state starts in S. Every allowed action maps a state in S back into S. Induction gives membership in S after every finite number of actions, and disjointness excludes B. □

The hypotheses of this theorem must be established from the chosen dynamics. They do not follow from exact simulation, a low seed cost, or an invertible change of representation.

Total transitions permit recursive continuation for every integer time. This is a mathematical existence statement. Control over a desired outcome, continuous-time continuation through a singularity, computational resources, and physical implementation are separate questions.

4 A precise example of representation invariance

For a Schwartz function $f : \mathbb{R} \to \mathbb{C}$, use the convention

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi ix\xi}\,dx.$$

Proposition 6 (Gaussian fixed point)**.** *For $g(x) = e^{-\pi x^2}$ one has $\mathcal{F}g = g$.*

Proof. Write $I(\xi) = (\mathcal{F}g)(\xi)$. Differentiation under the integral is justified by the integrable majorant $2\pi|x|e^{-\pi x^2}$. Since $g'(x) = -2\pi x g(x)$, integration by parts yields

$$I'(\xi) = i\int_{\mathbb{R}} g'(x)e^{-2\pi ix\xi}\,dx = -2\pi\xi I(\xi).$$

The boundary term vanishes by Gaussian decay. The Gaussian integral gives $I(0) = 1$, so the unique solution is $I(\xi) = e^{-\pi\xi^2}$. □

This is a fixed point of a representation transform. No singular time occurs in the argument. Fourier inversion preserves the mathematical signal; it does not, by itself, supply a dynamical continuation theorem.

5 Underdetermination and interpretation

Theorem 7 (A property is identifiable exactly when it factors)**.** *For a property $P_0 : X \to V$, there exists a function $\widehat{P} : \operatorname{im} R \to V$ with*

$$P_0 = \widehat{P} \circ R$$

if and only if P_0 is constant on every fiber of R.

Proof. If a factor exists, $R(x) = R(x')$ implies $P_0(x) = \widehat{P}(R(x)) = \widehat{P}(R(x')) = P_0(x')$. Conversely, fiber constancy makes $\widehat{P}(R(x)) := P_0(x)$ well defined. □

If two states have identical profiles but different values of P_0, those profiles alone cannot determine that property. This is underdetermination, not a logical contradiction. It does not establish that a particular hidden interpretation is true, nor that any imagined interpretation is consistent.

The mathematical conclusions concern observable identity, simulation fidelity, and explicitly assumed dynamics. They do not establish consciousness transfer, an afterlife, or human invulnerability.

Epilogue: the bridge. A bridge can symbolize the hope of meeting again. This is a personal interpretation, not a consequence of the theorems. Hope need not be presented as a mathematical guarantee.

References

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